Product Requirements Document (PRD)

Smart Faculty and Student Management System

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1.Executive Summary

1.1 Vision

The Smart Faculty and Student Management System (SFSMS) aims to build a next-generation, Agentic AI–powered College Management System that transforms traditional ERP platforms into intelligent, autonomous ecosystems capable of reasoning, predicting, and acting proactively.

The system seamlessly integrates academic, administrative, and operational workflows while empowering students, faculty, and administrators through real-time insights, automated decision support, and intelligent agents that continuously optimize institutional processes.

By shifting from reactive management to proactive intelligence, SFSMS envisions creating a smarter educational environment where routine operations are automated, risks are identified early, resources are optimized efficiently, and data-driven decisions enhance overall academic success and institutional productivity.

1.2 Problem Statement

Current college management systems primarily focus on data storage and basic process automation, but they often lack intelligence, adaptability, and proactive decision support. As institutions grow in scale and complexity, manual workflows and disconnected systems create operational inefficiencies and delays in academic decision-making. The following key challenges highlight the major pain points that the proposed Agentic AI–driven system aims to address.

Pain Point	**Description**
Manual Administrative Load	Faculty and administrative staff spend significant time on repetitive tasks such as timetable management, attendance tracking, and report preparation, reducing focus on academic quality.
Scattered Academic Data	Student records, attendance, performance metrics, and faculty information are stored across multiple systems, making access, analysis, and decision-making inefficient.
Reactive Decision Making	Institutions often identify issues like low attendance or poor academic performance only after outcomes are affected, limiting timely intervention.
Scheduling Conflicts	Manual handling of class schedules, exams, and room allocation frequently leads to clashes, resource underutilization, and last-minute changes.
Limited Intelligent Insights	Existing ERP systems primarily store data but lack AI-driven analytics and proactive recommendations to guide faculty, students, and administrators.

1.3 Solution Overview

A multi-agent AI-driven ERP system:

1. Faculty Agent — workload optimization, timetable intelligence
2. Student Success Agent — performance monitoring and intervention
3. Attendance Intelligence Agent — behavioral analytics
4. Scheduling Agent — autonomous conflict-free planning
5. Knowledge Agent — AI-powered document search
6. Finance Agent — smart fee tracking and prediction

1.4 Expected Impact

Metric	**Target**
Admin workload reduction	40-60%
Scheduling conflicts	Near Zero
Early risk detection	>80% accuracy
Faculty decision support	Real – Time insights

2.Product Objectives & Success Metrics

2.1 Primary Objectives

Objective	**KPI**	**Target**	**Measurement**
Automate admin workflows	Manual Task Reduction	>50%	System Logs
Improve student outcomes	Early risk detection	>80%	ML evaluation
Smart scheduling	Conflict Rate	<2%	Schedule Reports
Centralized intelligence	Data access time	Real - Time	Dashboard Analytics

2.2 Secondary Objectives

- Scalability for large institutions
- Modular agent architecture
- Role-based access control
- AI explainability for decisions

3. User Personas

Primary Users

Students

- Need academic guidance
- Require reminders and progress tracking
- Want centralized access to records

Faculty

- Need workload balancing
- Require class performance analytics
- Need quick access to student insights

Secondary Users

Admin Staff

- Manage schedules, fees, documents
- Require automation support

Institution Leadership

- Need predictive analytics
- Want data-driven decisions

4. Functional Requirements

4.1 Faculty Module (Faculty Agent)
Core Features

- Profile management
- AI workload allocation
- Leave prediction & substitute suggestions
- Performance analytics

Agent Behavior

- Detect overload patterns
- Recommend workload balancing

4.2 Student Module (Student Success Agent)
Features

- Academic records
- Progress tracking
- Personalized recommendations
- Risk prediction alerts

Agent Actions

- Detect declining performance
- Suggest study plans
- Notify mentors automatically

4.3 Smart Attendance System (Attendance Agent)
Features

- Biometric/RFID attendance
- Automated alerts
- Engagement analytics

Agent Intelligence

- Proxy detection probability
- Attendance-performance correlation
- Behavioral trend analysis

4.4 AI Organizer (Scheduling Agent)
Features

- Timetable generation
- Exam scheduling
- Room allocation

Agent Actions

- Conflict detection & auto-resolution
- Dynamic rescheduling
- Pattern learning for better scheduling

4.5 Document Management (Knowledge Agent)
Features

- AI-powered semantic search
- Version control

- Circular & syllabus repository

Agent Capabilities

- Natural language querying
- Context-aware document retrieval

4.6 Smart Fee Collection (Finance Agent)
Features

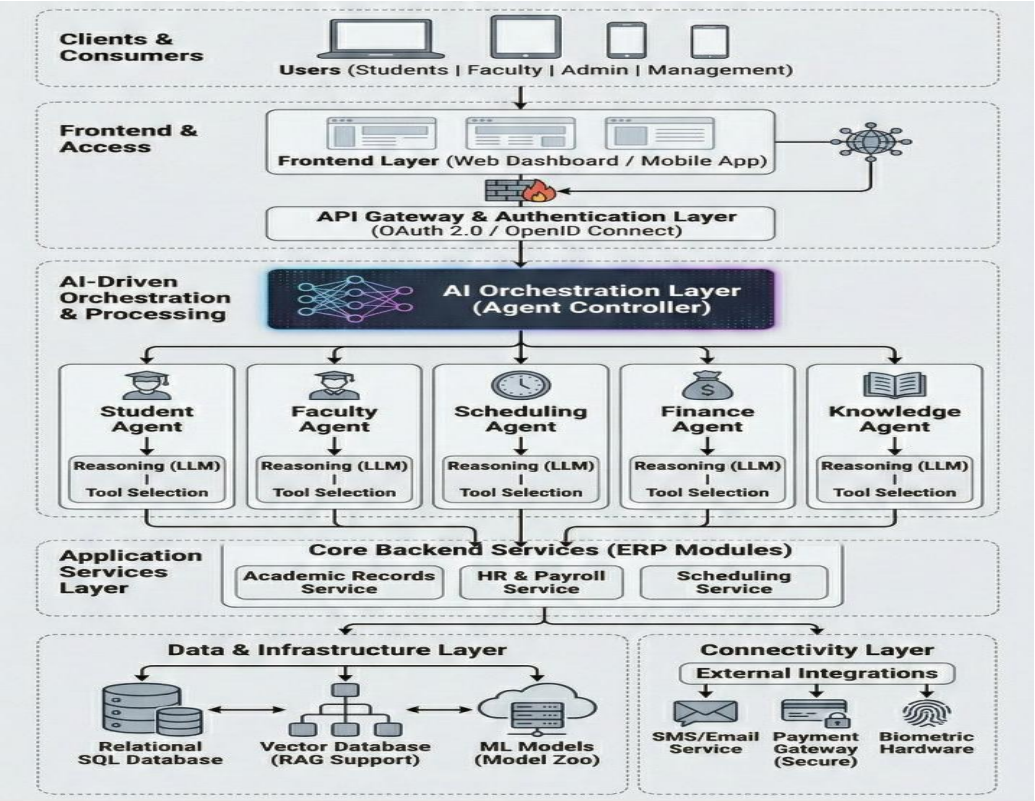
- Payment tracking
- Automated reminders
- Scholarship adjustment

Agent Intelligence

- Defaulter prediction
- EMI recommendation
- Payment behavior analytics

5. Technical Requirements

5.1 Architecture Overview



5.2 Technology Stack

Component	Technology	Reason
Backend	Django	Scalable APIs
AI Agents	LangChain / n8n	Agent orchestration
LLM	Llama / Mistral / GPT/Gemini	Reasoning capability
Database	MySQL	Structured ERP data
VecotorDB	FAISS/ChromaDB	Document search
ML Models	XGBoost / RandomForest	Prediction tasks
Frontend	Html, CSS, JS, React	Modern UI

5.3 Integrations

Mandatory:

- Student Information System (SIS)
- SMS/Email notification service
- Payment gateway

Optional:

- LMS integration
- Video conferencing APIs

6. Non-Functional Requirements

Performance

- Dashboard load < 3 seconds
- Real-time analytics updates
- Support 5,000+ concurrent users

Scalability

- Modular agent deployment
- Horizontal scaling

Reliability

- Daily backups
- Fault-tolerant architecture

7. User Experience Requirements

Dashboard Design Principles

- Role-based dashboards
- AI recommendation widgets
- One-click report export

Agent Interaction Style

- Professional but helpful tone
- Explainable AI recommendations

8. Implementation Roadmap

Phase 1 — Core ERP

- Student & Faculty modules
- Attendance system
- Basic dashboard

Phase 2 — Agent Integration

- Student Success Agent
- Scheduling Agent
- Knowledge Agent

Phase 3 — Intelligence Layer

- Predictive analytics
- Risk detection
- Autonomous recommendations

Phase 4 — Optimization

- Multi-agent collaboration
- Continuous learning system

9. Risk Assessment & Mitigation

Risk	Impact	Mitigation
AI incorrect suggestions	Medium	Human approval loop
Data privacy	High	Encryption + RBAC
Integration issues	Medium	API abstraction layer
High system load	Medium	Auto-scaling

10. Budget Estimate

Category	Estimate
Development	Team effort
Cloud Hosting	Medium
AI Models	Open-source preferred
Infrastructure	Minimal

11. Appendix

11.1 Glossary

Term	Definitions
Agentic AI	An AI system capable of autonomous decision-making, reasoning, and action execution based on defined goals and contextual data.
ERP(Enterprise Resource Planning)	A centralized system used to manage institutional operations such as academics, finance, and administration.
LLM(Large Language Model)	An AI model used for understanding, generating, and reasoning over natural language.
RAG(Retrieval Augmented Generation)	An AI approach that combines document retrieval with language generation for accurate, context-aware responses.
KPI(Key Performance Indicator)	Measurable value used to evaluate system success and performance outcomes.
RBAC(Role-based Access Control)	Security model that restricts system access based on user roles.

11.2 Assumptions

- Institutional data will be available in structured format.
- Users will have role-based access (Student, Faculty, Admin).
- Internet connectivity will be available for cloud-based deployment.
- Initial AI models will be trained on historical academic data.

11.3 Dependencies

- Database infrastructure (MySQL)
- AI model integration (LLM + ML models)
- Notification services (Email/SMS)
- Payment gateway integration (for fee module)

11.4 Future Enhancement

- Integration with Learning Management Systems (LMS)
- Advanced predictive analytics using deep learning
- Cross-institution performance benchmarking
- Mobile application deployment
- Continuous learning AI agents with feedback loops